Coordinate Geometry

Chapter 5 · Original slide 4 · Let's start simple…What is the equation of this line?And more importantly, why is it that?



Let's start simple...

What is the equation of this line?

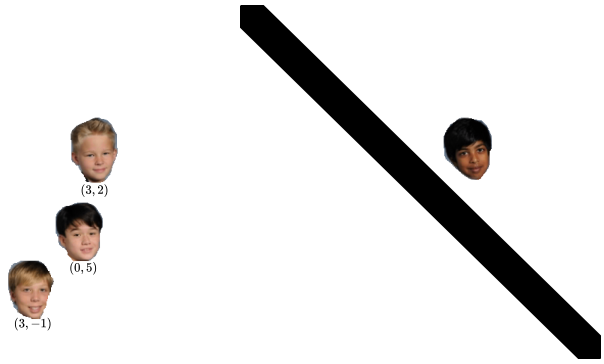
And more importantly, why is it that?

4
3
2
1
-1
-2
-3
-4

x -5 -4 -3 -2 -1 0 1 2 3 4 5 6

For any point we pick on the line, the x value is always 2.

Chapter 5 · Original slide 5 · Lines and Equations of Lines



! A line consists of all points which satisfy some equation in terms of x and/or y.

Anirudh's Line Club

Membership Rules:
$2x + y = 5$

Chapter 5 · Original slide 6 · Examples

This means we can substitute the values of a coordinate into our equation whenever we know the point lies on the line.

Find the coordinate of the point where the line $2x + y = 5$ cuts the x-axis.

The point $(5, a)$ lies on the line with equation $y = 3x + 2$. Determine the value of a.

Chapter 5 · Original slide 7 · Test Your Understanding

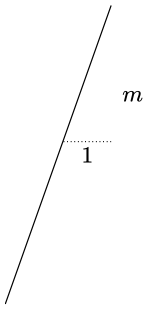
Determine where the line $x + 2y = 3$ crosses the:

a) y-axis:

b) x-axis:

What mistakes do you think it's easy to make?

Chapter 5 · Original slide 8 · Recap of gradient



The steepness of a line is known as the gradient.

It tells us what y changes by as x increases by 1.

So if the y value increased by 6 as the x value increased by 2, what is y increasing by for each unit increase of x?

How would that give us a suitable formula for the gradient m?

$m = \frac{\Delta y}{\Delta x}$

Δ is the (capital) Greek letter "delta" and means "change in".

Textbook Note:

The textbook uses $m = \frac{y_2 - y_1}{x_2 - x_1}$ for two points (x_1, y_1) and (x_2, y_2).

Reasons I don't use it for non-algebraic coordinates:

• Students often get the y_1 and y_2 the wrong way round (or with the x's)

• Students often make sign errors when dealing with negatives, e.g. $(-3) - (-4)$

• It can't be done as easily mentally,

• Students see it as "yet another formula to learn" when really all you need is to appreciate is what gradient is, i.e. "y change per x change".

Chapter 5 · Original slide 9 · Examples

Find the gradient of the line that goes through the points:

1

$(1, 4)(3, 10)$

2

$(5, 7)(8, 1)$

3

$(2, 2)(-1, 10)$

4

Show that the points $\mathrm{A}(3, 4), B(5, 5), C(11, 8)$ all lie on a straight line.

If points are 'collinear' they lie on the same line.

Chapter 5 · Original slide 10 · Further Example

The line joining $(2, -5)$ to $(4, a)$ has gradient -1.
Work out the value of a.

Chapter 5 · Original slide 11 · y=mx+c

One form we can put a straight line equation in is:

$y = mx + c$

Gradient

y-intercept

Why does it work?

• The $\mathbf{y}$-intercept by definition is the $\mathbf{y}$ value when $\mathbf{x=0}$. Substituting:
$\mathbf{y=(m\times 0)+c=c}$ as expected.

• By the definition of gradient, if we increase $\mathbf{x}$ by 1, $\mathbf{y}$ should increase by $\mathbf{m}$:
$\mathbf{y=m\,(x+1)+c=mx+m+c}$
which indeed has increased by $\mathbf{m}$.

Chapter 5 · Original slide 12 · y=mx+c

Determine the gradient and y-intercept of the line with equation $4x - 3y + 5 = 0$

Make y the subject so we have the form

$y = mx + c$

Put y on the side it's positive.

Divide each term by 3; don't write $y = \frac{4x+5}{3}$ otherwise it's not in the form $y = mx + c$

This is algebra, so use improper fractions, and not mixed numbers or recurring decimals.

Chapter 5 · Original slide 13 · ax+by+c=0

At GCSE, $\mathbf{y=mx+c}$ was the main form you would express a straight line equation, sometimes known as the 'slope-intercept form'.

But another common form is $\mathbf{ax+by+c=0}$, where a, b, c are integers. This is known as the 'standard' form.

Express $y = \frac{1}{3}x - \frac{2}{3}$ in the form $ax + by + c = 0$, where a, b, c are integers.

We don't want fractions, so multiply by an appropriate number.

Put everything on either side of equation. $-x + 3y + 2 = 0$ would also be OK.

We'll see on the next slide WHY we might want to put an equation in this form over $y = mx + c$...

Chapter 5 · Original slide 14 · Just for your interest…

Why might we want to put a straight line equation in the form $ax + by + c = 0$?

$2x + y = 4$



$y = mx + c$

"Slope-Intercept Form"

$ax + by + c = 0$

"Standard Form"

Coverage

$y = mx + c$ doesn't allow you to represent vertical lines. Standard form allows us to do this by just making b zero.

Usefulness

This more 'elegant' form also means it ties in with vectors and matrices. In FM, you will learn about the 'dot product' of two vectors:

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = ax + by$$

thus since $ax + by + c = 0$, we can represent a straight line using:

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} + c = 0 (1)$$

Symmetry

In general, the 'linear combination' of two variables x and y is $ax + by$, i.e. "some amount of x and some amount of y". There is a greater elegance and symmetry to this form over $y = mx + c$ because x and y appear similarly within the expression.

We can extend to 3D points to get the equation of a plane:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} + d = 0 (2)$$

Conveniently, in equation (1), the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is perpendicular to the line. And in equation (2), the vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is perpendicular to the plane. Nice!

$x + 4 = 0$

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = 4$$

$\begin{pmatrix} 2 \\ 1 \end{pmatrix}$

Chapter 5 · Original slide 15 · Test Your Understanding

Express $y = \frac{2}{5}x + \frac{3}{5}$ in the form $ax + by + c = 0$, where a, b, c are integers.

Chapter 5 · Original slide 16 · Exercise 5A – Pure Y1 Page 91

10 Show that the points $A(-2a, 5a)$, $B(0, 4a)$ and points $C(6a, a)$ are collinear. (3 marks)

9 Show that the points $A(2, 3)$, $B(4, 4)$ and $C(10, 7)$ can be joined by a straight line.

Chapter 5 · Original slide 17 · Exercise 5B – Pure Y1 Page 92-93

8 A line l cuts the x-axis at $(5, 0)$ and the y-axis at $(0, 2)$.

a Find the gradient of the line.

b Find an equation of the line in the form $ax + by + c = 0$.

10 The line l with gradient 3 and y-intercept $(0, 5)$ has the equation $ax - 2y + c = 0$.

Find the values of a and c. (2 marks)

9 Show that the line with equation $ax + by + c = 0$ has gradient $-\frac{a}{b}$ and cuts the y-axis at $-\frac{c}{b}$.

Chapter 5 · Original slide 18 · Exercise 5B – Pure Y1 Page 92-93

11 The straight line l passes through $(0, 6)$ and has gradient -2. It intersects the line with equation $5x - 8y - 15 = 0$ at point P. Find the coordinates of P.

12 The straight line l_1 with equation $y = 3x - 7$ intersects the straight line l_2 with equation $ax + 4y - 17 = 0$ at the point $P(-3, b)$.

a Find the value of b.

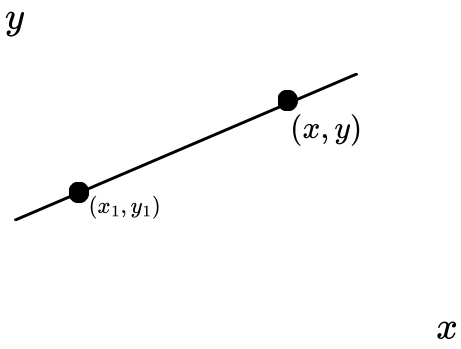
b Find the value of a.

Chapter 5 · Original slide 19 · Equations using two points/point + gradient

Find the equation of the line that goes through $(3, 5)$ and has gradient 2.

How would you have done this at GCSE?

Chapter 5 · Original slide 20 · A new way…



Suppose that (x_1, y_1) is some fixed point on the line that we specify (e.g. $(3, 5)$).

Suppose that (x, y) represents a generic point on the line, which is allowed to change as we consider different points on this line.

Then:

$m = \frac{y-y_1}{x-x_1}$

Thus:

Notes: Note that x_1 and y_1 are constants while x and y are variables. The latter are variables because as these 'vary', we get different points on the line.

! The equation of a line that has gradient m and passes through a point (x_1, y_1) is:

$y - y_1 = m(x - x_1)$

Let's revisit:

Find the equation of the line that goes through $(3, 5)$ and has gradient 2.

No manipulation required! Just plug in numbers and we're done.

Again, note that x and y are variables so need to remain as such.

Chapter 5 · Original slide 21 · Quickfire Questions

In a nutshell: You can use this formula whenever you have (a) a gradient and (b) any point on the line.

Gradient	Point	(Unsimplified) Equation
3	$(1, 2)$	
5	$(3, 0)$	
2	$(-3, 4)$	
$\frac{1}{2}$	$(1, -5)$	
9	$(-4, -4)$	

Important Side Note: I've found that many students shun this formula and just use the GCSE method. Please persist with it – it'll be much easier when fractions are involved.

Further Mathematicians, don't even think about using the GCSE method, because you'll encounter massive headaches when you consider algebraic points. Trust me on this one!

Chapter 5 · Original slide 22 · Using 2 points

Find the equation of the line that goes through $(4, 5)$ and $(6, 2)$, giving your equation in the form $ax + by + c = 0$.

We need to work out the gradient first before we can use $y - y_1 = m(x - x_1)$:

Test Your Understanding:

Find the equation of the line that goes through $(-1, 9)$ and $(4, 5)$, giving your equation in the form $ax + by + c = 0$.

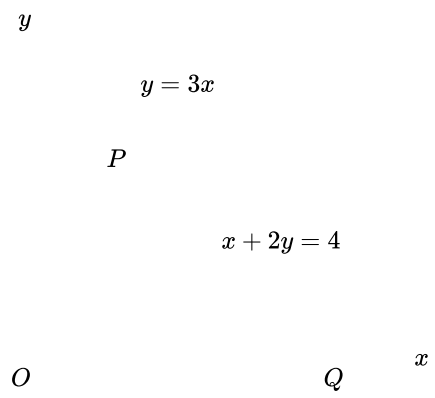
Chapter 5 · Original slide 23 · Exercise 5C – Pure Y1 Page 94-95

3 Find the equation of the line l which passes through the points $A(7, 2)$ and $B(9, -8)$. Give your answer in the form $ax + by + c = 0$.

6 The straight line l passes through $(7a, 5)$ and $(3a, 3)$. An equation of l is $x + by - 12 = 0$. Find the value of a and the value of b.

5 The straight line l passes through $(a, 4)$ and $(3a, 3)$. An equation of l is $x + 6y + c = 0$. Find the value of a and the value of c.

Chapter 5 · Original slide 24 · Intersection of lines



The diagram shows two lines with equations $y = 3x$ and $x + 2y = 4$, which intersect at the point P.

a) Determine the coordinates of P.

We know from Chapter 4 we can just solve two equations simultaneously.

b) The line $x + 2y = 4$ intersects the x-axis at the point Q. Determine the coordinate of Q.

Chapter 5 · Original slide 25 · Test Your Understanding

The straight line L_1 passes through the points $(-1, 3)$ and $(11, 12)$.
(a) Find an equation for L_1 in the form $ax + by + c = 0$, where a, b and c are integers. (4)
The line L_2 has equation $3y + 4x - 30 = 0$.
(b) Find the coordinates of the point of intersection of L_1 and L_2. (3)

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Chapter 5 · Original slide 26 · Exercise 5D – Pure Y1 Page 96

6 The line that passes through the points $(-3, -5)$ and $(4, 9)$ meets the y-axis at the point G. Work out the coordinates of the point G.

9 The lines $y = 4x - 10$ and $y = x - 1$ intersect at the point T. Find the equation of the line with gradient $-\frac{2}{3}$ that passes through the point T. Write your answer in the form $ax + by + c = 0$, where a, b and c are integers.

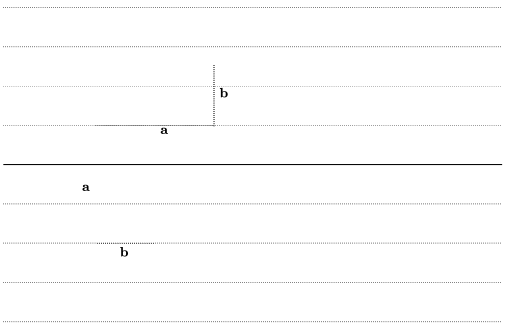
Chapter 5 · Original slide 27 · Exercise 5D – Pure Y1 Page 96

14 The line $y = 4x + 5$ meets the y-axis at the point C. The line $y = -3x - 15$ meets the x-axis at the point D. Find the equation of the line joining the points C and D. Write your answer in the form $ax + by + c = 0$, where a, b and c are integers.

16 The lines $y = -2x + 1$ and $y = x + 7$ intersect at the point L. The point M has coordinates $(-3, 1)$. Find the equation of the line that passes through the points L and M.

Coordinate Geometry

Chapter 5 · Original slide 28 · y



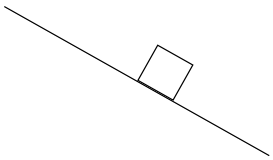
4
3
2
1
-1
-2
-3
-4

x -5 -4 -3 -2 -1 0 1 2 3 4 5 6

Using the changes above, we can see the gradient of one line is $\mathbf{m=\frac{b}{a}}$ and the other $\mathbf{m=-\frac{a}{b}}$. One is the 'negative reciprocal' of the other.

Find the gradients of each pair of perpendicular lines. What do you notice?

Chapter 5 · Original slide 29 · Perpendicular Lines



The gradients of parallel lines are equal.

If two lines are perpendicular, then the gradient of one is the negative reciprocal of the other.

$m_1 = -\frac{1}{m_2}$

To show that two lines are perpendicular:

$m_1 m_2 = -1$

Gradient	Gradient of Perpendicular Line
2	
-3	
$\frac{1}{4}$	
5	
$-\frac{2}{7}$	
$\frac{7}{5}$	

Chapter 5 · Original slide 30 · Example Problems

1 2 3

A line is goes through the point (9,10) and is perpendicular to another line with equation $y = 3x + 2$. What is the equation of the line?

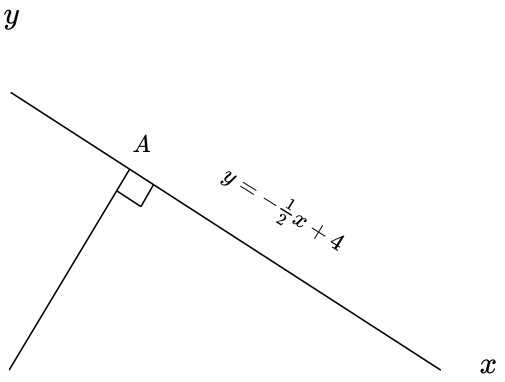
A line L_1 goes through the points $A(1, 3)$ and $B(3, -1)$. A second line L_2 is perpendicular to L_1 and passes through point B. Where does L_2 cross the x-axis?

Are the following lines parallel, perpendicular, or neither?

$y = \frac{1}{2}x$

$2x - y + 4 = 0$

Chapter 5 · Original slide 31 · Test Your Understanding



2

Determine the point A.

1

A line goes through the point (4,7) and is perpendicular to another line with equation y=2x+2. What is the equation of the line? Put your answer in the form ax+by+c=0, where a,b,c are integers.

Chapter 5 · Original slide 32 · Exercise 5E/5F – Pure Y1 Page 96

3 The coordinates of a quadrilateral $ABCD$ are $A(-6, 2), B(4, 8), C(6, 1)$ and $D(-9, -8)$. Show that the quadrilateral is a trapezium.

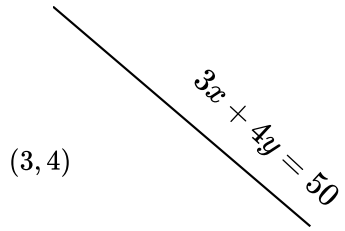
7 A line is parallel to the line $2x - 3y - 1 = 0$ and it passes through the point $(0, 0)$. Write down the equation of the line.

Chapter 5 · Original slide 33 · Exercise 5F – Pure Y1 Page 100

9 The line l passes through the points $(-3, 0)$ and $(3, -2)$ and the line n passes through the points $(1, 8)$ and $(-1, 2)$. Show that the lines l and n are perpendicular.

10 The vertices of a quadrilateral $ABCD$ have coordinates $A(-1, 5), B(7, 1), C(5, -3)$ and $D(-3, 1)$. Show that the quadrilateral is a rectangle.

Chapter 5 · Original slide 34 · Exercise 5E/5F – Pure Y1 Page 96



Extension Problems

2

[MAT 2014 1D] The reflection of the point $(1, 0)$ in the line $y = mx$ has coordinates: (in terms of m)

1

[MAT 2004 1D]

What is the reflection of the point $(3, 4)$ in the line $3x + 4y = 50$?

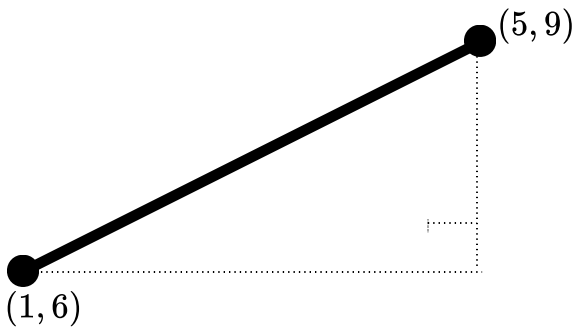
3

[STEP I 2004 Q6] The three points A, B, C have coordinates $(p_1, q_1), (p_2, q_2)$ and (p_3, q_3), respectively. Find the point of intersection of the line joining A to the midpoint of BC, and the line joining B to the midpoint of AC. Verify that this point lies on the line joining C to the midpoint of AB.

The point H has coordinates $(p_1 + p_2 + p_3, q_1 + q_2 + q_3)$. Show that if the line AH intersects the line BC at right angles, then $p_2^2 + q_2^2 = p_3^2 + q_3^2$, and write down a similar result if the line BH intersects the line AC at right angles.

Deduce that if AH is perpendicular to BC and also BH is perpendicular to AC, then CH is perpendicular to AB.

Chapter 5 · Original slide 35 · Distances between points



Recall: Δ (said 'delta') means "change in".

How could we find the distance between these two points?

Hint: Pythagoras

Form a right-angled triangle using the change in x and change in y, then use Pythagoras.

! Distance between two points:

$$\sqrt{(\Delta x)^2 + (\Delta y)^2}$$

Chapter 5 · Original slide 36 · Examples

Distance between:

(3,4) and $(5, 7)$

(5,1) and $(6, -3)$

(0,-2) and $(-1, 3)$

Note: Unlike with gradient, we don't care if the difference is positive or negative (it's being squared to make it positive anyway!)

Quickfire Questions:

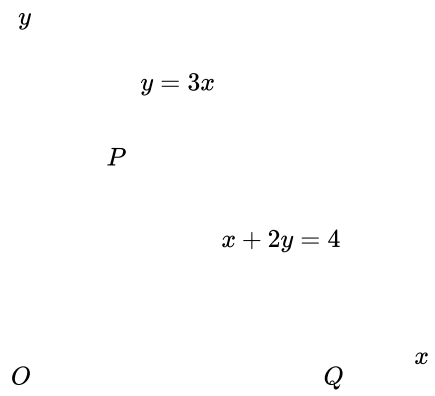
Distance between:

(1,10) and $(4, 14)$

(3,-1) and $(0, 1)$

(-4,-2) and $(-12, 4)$

Chapter 5 · Original slide 37 · Area of Shapes



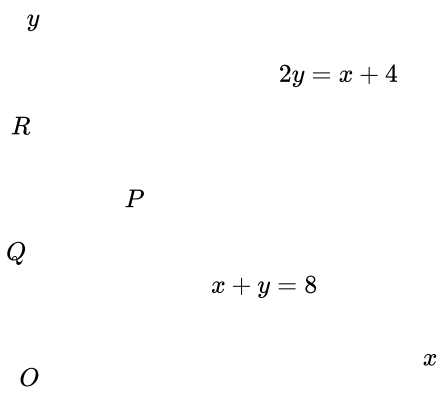
The diagram shows two lines with equations $y = 3x$ and $x + 2y = 4$, which intersect at the point P.

a) Determine the coordinates of P.

(We did this in a previous lesson)

b) The line $x + 2y = 4$ intersects the x-axis at the point Q. Determine the area of the triangle OPQ.

Chapter 5 · Original slide 38 · Further Example



Tip: When finding areas of triangles in exam questions, one line is often vertical or horizontal. You should generally choose this to be the 'base' of your triangle.

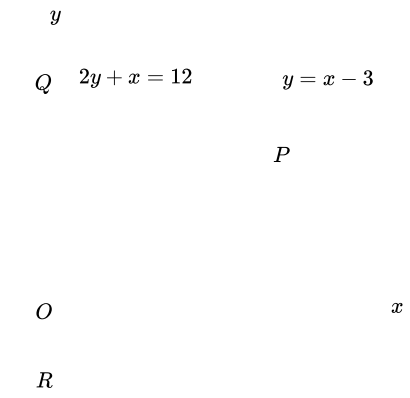
a

Determine the length of PQ.

b

Determine the area PQR.

Chapter 5 · Original slide 39 · Test Your Understanding



a) Determine the coordinate of P.

b) Determine the area of PQR.

c) Determine the length PQ.

Chapter 5 · Original slide 40 · Exercise 5G – Pure Y1 Page 102

7 A point P lies on the line with equation $y = 4 - 3x$. The point P is a distance $\sqrt{34}$ from the origin. Find the two possible positions of point P. (5 marks)

8 The vertices of a triangle are $A(2, 7)$, $B(5, -6)$ and $C(8, -6)$.
a Show that the triangle is a scalene triangle.
b Find the area of the triangle ABC.

Chapter 5 · Original slide 41 · Exercise 5G – Pure Y1 Page 103

12 The straight line l_1 passes through the point $(-4, 14)$ and has gradient $-\frac{1}{4}$.
a Find an equation for l_1 in the form $ax + by + c = 0$, where a, b and c are integers.
b Write down the coordinates of A, the point where straight line l_1 crosses the y-axis.
The straight line l_2 passes through the origin and has gradient 3. The lines l_1 and l_2 intersect at the point B.
c Calculate the coordinates of B.
d Calculate the exact area of $\triangle OAB$.

Chapter 5 · Original slide 42 · Exercise 5G – Pure Y1 Page 102-103

Extension Problems

1

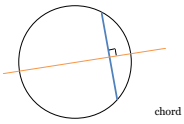
[MAT 2001 1C]

The shortest distance from the origin to the line $3x + 4y = 25$ is what?

$3x + 4y = 25$

$(0, 0)$

Chapter 6 · Original slide 4 · Midpoints and Perpendicular Bisectors



Later in the chapter you will need to find the perpendicular bisector of a chord of a circle.



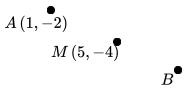
What two properties does a perpendicular bisector of two points **A** and **B** have?

perpendicular bisector

$A\,(2, 5)$

Just find the mean of the x values and the mean of the y values.

Chapter 6 · Original slide 5 · Test Your Understanding



Find the perpendicular bisector of the line AB where A and B have the coordinates:

a) $A\,(4, 7), B\,(10, 17)$

b) $A\,(x_1, y_1), B\,(x_2, y_2)$

A line segment AB is the diameter of a circle with centre $(5, -4)$. If A has coordinates $(1, -2)$, what are the coordinates of B?

a b

$M\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$m_{AB} = \frac{y_2-y_1}{x_2-x_1}$ $\quad$ $m_{\perp} = \frac{x_1-x_2}{y_2-y_1}$

$y - \frac{y_1+y_2}{2} = \frac{x_1-x_2}{y_2-y_1}\left(x - \frac{x_1+x_2}{2}\right)$

We could use the formula $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$, but it is easy to see from the diagram that whatever movement there is from A to M, we can continue to get to B.

Note: Do not try to memorise this!

Chapter 6 · Original slide 6 · Exercise 6A - Y1 Page 115

6 The line segment JK is a diameter of a circle, where J and K are $(\frac{3}{4}, \frac{4}{3})$ and $(-\frac{1}{2}, 2)$ respectively. The centre of the circle lies on the line segment with equation $y = 8x + b$. Find the value of b.

7 The line segment AB is a diameter of a circle, where A and B are $(0, -2)$ and $(6, -5)$ respectively. Show that the centre of the circle lies on the line $x - 2y - 10 = 0$.

Chapter 6 · Original slide 7 · Exercise 6B - Y1 Page 116

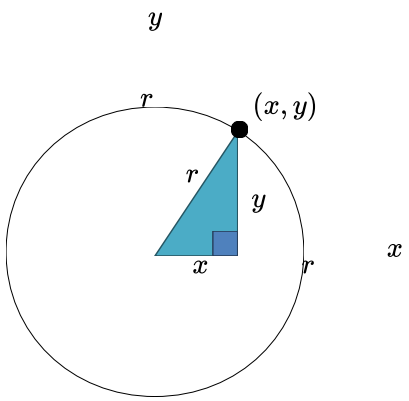
2 The line FG is a diameter of the circle centre C, where F and G are $(-2, 5)$ and $(2, 9)$ respectively. The line l passes through C and is perpendicular to FG. Find the equation of l. (7 marks)

3 The line JK is a diameter of the circle centre P, where J and K are $(0, -3)$ and $(4, -5)$ respectively. The line l passes through P and is perpendicular to JK. Find the equation of l. Write your answer in the form $ax + by + c = 0$, where a, b and c are integers.

Chapter 6 · Original slide 8 · Exercise 6B - Y1 Page 117

5 Point X has coordinates $(7, -2)$ and point Y has coordinates $(4, q)$. The perpendicular bisector of XY has equation $y = 4x + b$. Find the value of q and the value of b.

Chapter 6 · Original slide 9 · Equation of a circle



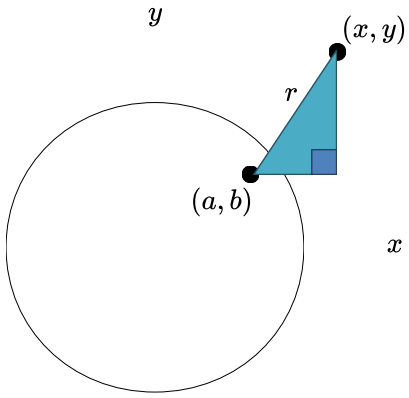
Recall that a line can be a set of points (x, y) that satisfy some equation.

Suppose we have a point (x, y) on a circle centred at the origin, with radius r. What equation must (x, y) satisfy?

(Hint: draw a right-angled triangle inside your circle, with one vertex at the origin and another at the circumference)

$x^2 + y^2 = r^2$

Chapter 6 · Original slide 10 · Equation of a circle



Now suppose we shift the circle so it's now centred at (a, b). What's the equation now?

(Hint: What would the sides of this right-angled triangle be now?)

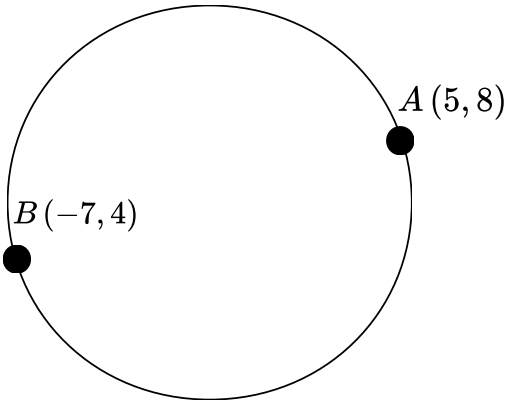
! The equation of a circle with centre (a, b) and radius r is:

$(x - a)^2 + (y - b)^2 = r^2$

Chapter 6 · Original slide 11 · Quickfire Questions

Centre	Radius
Equation	$(0,0)$
$(x+3)^2+(y-5)^2=1$	$(x+5)^2+(y-2)^2=49$
$(x+6)^2+y^2=16$	$(x-1)^2+(y+1)^2=3$
$(x+2)^2+(y-3)^2=8$	

Chapter 6 · Original slide 12 · Finding the equation using points



A line segment AB is the diameter of a circle, where A and B have coordinates $(5,8)$ and $(-7,4)$ respectively. Determine the equation of the circle.

Hint: What two things do we need to use the circle formula?

Chapter 6 · Original slide 13 · Test Your Understanding

The points A and B have coordinates $(5, -1)$ and $(13, 11)$ respectively.
(a) Find the coordinates of the mid-point of AB. (2)
Given that AB is a diameter of the circle C,
(b) find an equation for C. (4)

Chapter 6 · Original slide 14 · Completing the square

When the equation of a circle is in the form $(x-a)^2 + (y-b)^2 = r^2$, we can instantly read off the centre (a, b) and the radius r.

But what if the equation wasn't in this form?

Hint: Have we seen a method in a previous chapter that allows us to turn a x^2 term and a x term into a single expression involving x?

Complete the square!

Find the centre and radius of the circle with equation $x^2 + y^2 - 6x + 2y - 6 = 0$

Rearrange terms so that x terms are together and y terms are together.

Textbook Note: There's a truly awful method, initially presented in the textbook, that allows you to find the centre/radius without completing the square. Don't even contemplate using it.

Chapter 6 · Original slide 15 · Further Example

The circle C with centre T and radius r has equation
$x^2 + y^2 - 20x - 16y + 139 = 0$

(a) Find the coordinates of the centre of C. (3)
(b) Show that $r = 5$. (2)

Chapter 6 · Original slide 16 · Example

Given the generic equation of a circle $x^2 + y^2 + ax + by + c = 0$ find a formula to determine the co-ordinate of the centre and the radius of the circle.

$\left(x + \frac{a}{2}\right)^2 + \left(y + \frac{b}{2}\right)^2 = -c + \frac{a^2}{4} + \frac{b^2}{4}$

$r = \left(\frac{a}{2}\right)^2 + \left(\frac{b}{2}\right)^2 - c$

$c\left(-\frac{a}{2}, -\frac{b}{2}\right)$

Chapter 6 · Original slide 17 · Exercise 6C - Y1 Page 120

5 The line PQ is the diameter of the circle, where P and Q are $(5, 6)$ and $(-2, 2)$ respectively. Find the equation of the circle. (5 marks)

6 The point $(1, -3)$ lies on the circle $(x - 3)^2 + (y + 4)^2 = r^2$. Find the value of r.

Chapter 6 · Original slide 18 · Exercise 6C - Y1 Page 120

8 a Show that $x^2 + y^2 - 4x - 11 = 0$ can be written in the form $(x - a)^2 + y^2 = r^2$, where a and r are numbers to be found. (2 marks)
b Hence write down the centre and radius of the circle with equation $x^2 + y^2 - 4x - 11 = 0$. (2 marks)

9 a Show that $x^2 + y^2 - 10x + 4y - 20 = 0$ can be written in the form $(x - a)^2 + (y - b)^2 = r^2$, where a, b and r are numbers to be found. (2 marks)
b Hence write down the centre and radius of the circle with equation $x^2 + y^2 - 10x + 4y - 20 = 0$. (2 marks)

Chapter 6 · Original slide 19 · Exercise 6C - Y1 Page 120

12 The point $P(7, -14)$ lies on the circle with equation $x^2 + y^2 + 6x - 14y = 17$. The point Q also lies on the circle such that PQ is a diameter. Find the coordinates of point Q.

Chapter 6 · Original slide 20 · Extension

1

[MAT 2009 1B] The point on the circle $x^2 + y^2 + 6x + 8y = 75$ which is closest to the origin, is at what distance from the origin?

2

[MAT 2007 1D]

The point on the circle

$(x - 5)^2 + (y - 4)^2 = 4$

which is closest to the circle

$(x - 1)^2 + (y - 1)^2 = 1$

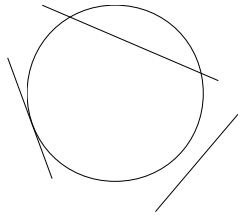
has what coordinates?

3

[MAT 2016 1I] Let a and b be positive real numbers. If

$x^2 + y^2 \leq 1$ then the largest that $ax + by$ can equal is what? Give your expression in terms of a and b.

Chapter 6 · Original slide 21 · Intersections of Lines and Circles



Recall that to consider the intersection of two lines, we attempt to solve them simultaneously by substitution, potentially using the discriminant to show that there are no solutions (and hence no points of intersection).

Show that the line $y = x + 3$ never intersects the circle with equation $x^2 + y^2 = 1$.

2 intersections (such a line is known as a secant of the circle)

1 intersections (such a line is known as a tangent of the circle)

0 intersections

Chapter 6 · Original slide 22 · Test Your Understanding

Find the points of intersection where the line $y = x + 6$ meets $x^2 + (y - 3)^2 = 29$.

Using an algebraic (and not geometric) method, determine the k such that the line $y = x + k$ touches the circle with equation $x^2 + y^2 = 1$.

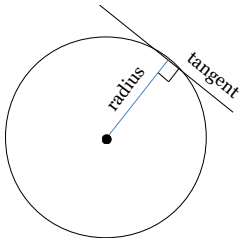
Chapter 6 · Original slide 23 · Exercise 6D - Y1 Page 122

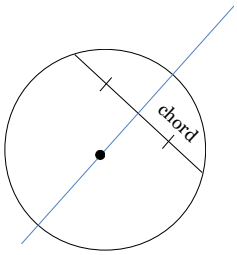
6 a Show that the line $x + y = 11$ meets the circle with equation $x^2 + (y - 3)^2 = 32$ at only one point.
(4 marks)
b Find the coordinates of the point of intersection.
(1 marks)

Chapter 6 · Original slide 24 · Exercise 6D - Y1 Page 122

11 The line with equation $y = 4x - 1$ does not intersect the circle with equation $x^2 + 2x + y^2 = k$.
Find the range of possible values of k.

Chapter 6 · Original slide 25 · Tangents, Chords, Perpendicular Bisectors





There are two circle theorems that are of particular relevance to problems in this chapter, the latter you might be less familiar with:

perpendicular bisector

The perpendicular bisector of any chord passes through the centre of the circle.

Why this will help:

The first thing we did in this chapter is find the equation of the perpendicular bisector. If we had two chords, and hence found two bisectors, we could find the point of intersection, which would be the centre of the circle.

The tangent is perpendicular to the radius (at the point of intersection).

Why this will help:

If we knew the centre of the circle and the point of intersection, we can easily find the gradient of the radius, and thus the gradient and hence equation of the tangent.

Chapter 6 · Original slide 26 · Examples



A circle C has equation
$(x-4)^2+(y+4)^2=10$

The line l is a tangent to the circle and has gradient -3. Find two possible equations for l, giving your answers in the form $y=mx+c$.

The circle C has equation
$(x-3)^2+(y-7)^2=100$.

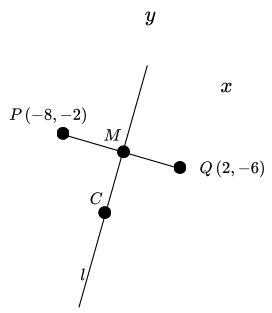
a) Verify the point $P\,(11,1)$ lies on C.

b) Find an equation of the tangent to C at the point P, giving your answer in the form $ax+by+c=0$

Tip: Use ‘subscripting’ of variables to make clear to the examiner (and yourself!) what you’re calculating.

This time we have the gradient, but don’t have the points where the tangent(s) intersect the radius.

Chapter 6 · Original slide 27 · Determining the Circle Centre



Exam Tip: If you're not asked for the equation in a particular form, just leave it as it is.

The points P and Q lie on a circle with centre C, as shown in the diagram. The point P has coordinates $(-8, -2)$ and the point Q has coordinates $(2, -6)$. M is the midpoint of the line segment PQ.

The line l passes through the points M and C.

a) Find an equation for l.

b) Given that the y-coordinate of C is -9:
i) show that the x-coordinate of C is -5.
ii) find an equation of the circle.

a b c

We know the line going through the midpoint of a chord and a centre of the circle is the perpendicular bisector of the chord.

Chapter 6 · Original slide 28 · Test Your Understanding

A circle has centre $C\,(3, 5)$, and goes through the point $P(6, 9)$. Find the equation of the tangent of the circle at the point P, giving your equation in the form $ax + by + c = 0$ where a, b, c are integers..

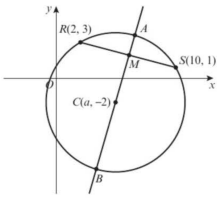
A circle passes through the points $A(0, 0)$ and $B\,(4, 2)$. The centre of the circle has x value -1. Determine the equation of the circle.

Chapter 6 · Original slide 29 · Exercise 6E - Y1 Page 126-128

6 The tangent to the circle $(x+4)^2+(y-1)^2=242$ at $(7,-10)$ meets the y-axis at S and the x-axis at T.
a Find the coordinates of S and T. (5 marks)
b Hence, find the area of $\triangle OST$, where O is the origin. (3 marks)

Chapter 6 · Original slide 30 · Exercise 6E - Y1 Page 126-128

The points R and S lie on a circle with centre $C(a, -2)$, as shown in the diagram.
The point R has coordinates (2, 3) and the point S has coordinates (10, 1).
M is the midpoint of the line segment RS.
The line l passes through M and C.
a Find an equation for l. **(4 marks)**
b Find the value of a. **(2 marks)**
c Find the equation of the circle. **(3 marks)**



d Find the points of intersection, A and B, of the line l and the circle. (5 marks)

10 The points R and S lie on a circle with centre $C(a,-2)$, as shown in the diagram.
The point R has coordinates $(2,3)$ and the point S has coordinates $(10,1)$.
M is the midpoint of the line segment RS.
The line l passes through M and C.
a Find an equation for l. (4 marks)
b Find the value of a. (2 marks)
c Find the equation of the circle. (3 marks)

Chapter 6 · Original slide 31 · Exercise 6E - Y1 Page 126-128

Extension:

1

[MAT 2012 1A] Which of the following lines is a tangent to the circle with equation

$x^2 + y^2 = 4$?

A) $x + y = 2$

B) $y = x - 2\sqrt{2}$

C) $x = \sqrt{2}$

D) $y = \sqrt{2} - x$

(This is not a tangent/chord question but is worthwhile regardless!)

2

[AEA 2006 Q4] The line with equation $y = mx$ is a tangent to the circle C_1 with equation $(x + 4)^2 + (y - 7)^2 = 13$

(a) Show that m satisfies the equation $3m^2 + 56m + 36 = 0$

The tangents from the origin O to C_1 touch C_1 at the points A and B.

(b) Find the coordinates of the points A and B.

Another circle C_2 has equation $x^2 + y^2 = 13$. The tangents from the point $(4, -7)$ to C_2 touch it at the points P and Q.

(c) Find the coordinates of either the point P or the point Q.

Mark scheme on next slide.

3

[STEP 2005 Q6]

(i) The point A has coordinates $(5, 16)$ and the point B has coordinates $(4, -4)$. The variable P has coordinates (x, y) and moves on a path such that $AP = 2BP$. Show that the Cartesian equation of the path of P is $(x + 7)^2 + y^2 = 100$.

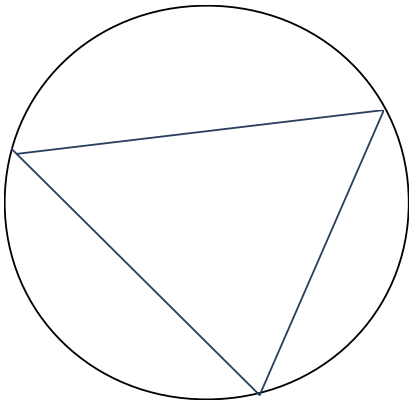
(ii) The point C has coordinates $(a, 0)$ and the point D has coordinates $(b, 0)$. The variable point Q moves on a path such that $QC = k \times QD$, where $k > 1$.
Given that the path of Q is the same as the path of P, show that $\frac{a+7}{b+7} = \frac{a^2+51}{b^2+51}$
Show further that $(a + 7)(b + 7) = 100$, in the case $a \neq b$.

Mark scheme on next slide.

Chapter 6 · Original slide 32 · Mark Scheme for Extension Question 2

Chapter 6 · Original slide 33 · Mark Scheme for Extension Question 3

Chapter 6 · Original slide 34 · Triangles in Circles



We'd say:

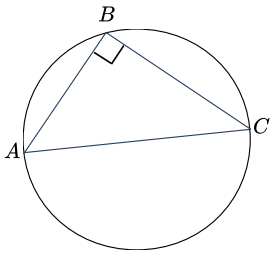
• The triangle inscribes the circle.
(A shape inscribes another if it is inside and its boundaries touch but do not intersect the outer shape)

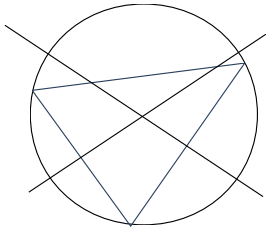
• The circle circumscribes the triangle.

• If the circumscribing shape is a circle, it is known as the circumcircle of the triangle.

• The centre of a circumcircle is known as the circumcentre.

Chapter 6 · Original slide 35 · Triangles in Circles





If $\angle ABC = 90^\circ$ then:

• $\mathbf{AC}$ is the diameter of the circumcircle of triangle $\mathbf{ABC}$.

Similarly if AC is the diameter of a circle:

• $\mathbf{\angle ABC = 90^\circ}$ therefore $\mathbf{AB}$ is perpendicular to $\mathbf{BC}$.

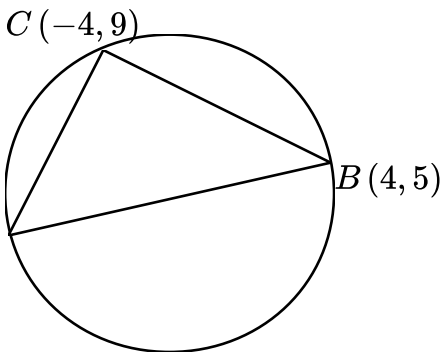
• $\mathbf{AB^2 + BC^2 = AC^2}$

Given three points/a triangle we can find the centre of the circumcircle by:

• Finding the equation of the perpendicular bisectors of two different sides.

• Find the point of intersection of the two bisectors.

Chapter 6 · Original slide 36 · Example



[Textbook] The points $A\,(-8, 1), B\,(4, 5), C\,(-4, 9)$ lie on a circle.

a) Show that AB is a diameter of the circle.

$A\,(-8, 1)$

Method 1:

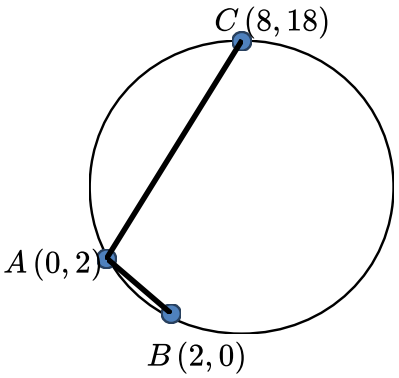
Show that $AC^2 + BC^2 = AB^2$

Method 2: (not in textbook!)

Show that AC is perpendicular to BC.

b) Hence find the equation of the circle.

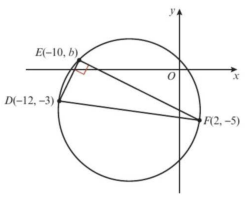
Chapter 6 · Original slide 37 · Example



The points $A\,(0,2), B\,(2,0), C\,(8,18)$ lie on the circumference of a circle. Determine the equation of the circle.

Chapter 6 · Original slide 38 · Exercise 6F - Y1 Page 131-132

8 The points $D(-12,-3), E(-10,b)$ and $F(2,-5)$ lie on the circle C as shown in the diagram.
Given that $\angle DEF = 90°$ and $b > 0$
a show that $b = 1$ (5 marks)
b find an equation for C. (4 marks)



Chapter 6 · Original slide 39 · Exercise 6F - Y1 Page 131-132

9 A circle has equation
$x^2 + 2x + y^2 - 24y - 24 = 0$.
a Find the centre and radius of the circle.
b The points $A(-13, 17)$ and $B(11, 7)$ both lie on the circumference of the circle. Show that AB is a diameter of the circle.
c The point C lies on the negative x-axis and the angle $ACB = 90°$. Find the coordinates of C.

Chapter 6 · Original slide 40 · Exercise 6F - Y1 Page 131-132

Extension:

1

[STEP 2009 Q8 Edited] If equation of the circle C is $(x - 2t)^2 + (y - t)^2 = t^2$, where t is a positive number, it can be shown that C touches the line $y = 0$ as well as the line $3y = 4x$.

Find the equation of the incircle of the triangle formed by the lines $y = 0, 3y = 4x$ and $4y + 3x = 15$.

Note: The incircle of a triangle is the circle, lying totally inside the triangle, that touches all three sides.